

Chile-ED-Resp: A Curated and Reproducible Weekly Hospital Dataset for Respiratory Emergency-Demand Forecasting in Chile

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Citation: Balcázar-Quimi, W.; Angulo, C.; Cornejo-Gaete, C.; Salinas-Neira, B.; Fernández-Bastías, J.; Tupac-Yupanqui, M.; Vidal-Silva, C. Chile-ED-Resp: A Curated Weekly Hospital Dataset for Respiratory Emergency-Demand Forecasting. *Data* **2026**, *1*, 0. <https://doi.org/>

Received:
Accepted:
Published:

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Abstract: Respiratory emergencies create recurrent seasonal pressure on hospital services, yet reproducible forecasting research is often hindered by the effort required to retrieve, interpret, harmonize, and document public administrative data. Chile-ED-Resp is a curated secondary dataset and reproducible processing pipeline derived from the official Chilean Ministry of Health/DEIS dataset “Atenciones de urgencias de causas respiratorias por semana epidemiológica”, produced by the Sistema de Atención Diaria de Urgencias (SADU). Version 1.1.0 uses the closed 2022–2025 period and was built from the official Parquet snapshot retrieved on 30 August 2026. After restriction to hospitals and hospital emergency units (UEH), the curated long artifact contains 449,844 rows and 32 variables, representing 180 hospitals and 12 official respiratory cause labels. A complementary forecasting artifact contains 37,620 hospital-week rows and 38 variables on a fixed 209-position statistical-week calendar (52 weeks in 2022–2024 and 53 in 2025); 178 of 180 hospitals (98.9%) satisfy the predefined $\geq 95\%$ target-coverage criterion. Acute upper respiratory infection is retained under its official Spanish label and acronym, IRA Alta. Missing source records remain missing and are distinguished from explicitly reported zeros. Precomputed target-derived predictors use outcomes available through the preceding week and are intended for rolling-origin one-week-ahead evaluation, not an unqualified fixed-origin forecast of the entire test year. Reproducibility is supported through source metadata, SHA-256 checksums, machine-readable descriptive audits, 15 validation checks, unit tests, and an example that recomputes predictors at each forecast origin. The resource provides a transparent foundation for forecasting, benchmarking, health-informatics teaching, and secondary analyses while preserving source provenance and licensing conditions.

Dataset: Chile-ED-Resp, version 1.1.0. Creator: Balcázar-Quimi, W.; Angulo, C.; Cornejo-Gaete, C.; Salinas-Neira, B.; Fernández-Bastías, J.; Tupac-Yupanqui, M.; Vidal-Silva, C. Publisher/location: Zenodo. Version-specific DOI: <https://doi.org/10.5281/zenodo.22209795>. Development repository: <https://github.com/cvidalmsu/chile-ed-resp-benchmark>.

Dataset License: Curated data adaptations: Creative Commons Attribution-NonCommercial 2.0 Generic (CC BY-NC 2.0), requiring non-commercial use, source attribution, identification of changes, and retention of the license notice; software: MIT License.

Keywords: emergency department; respiratory illness; hospital demand; time series; forecasting; data curation; DEIS; SADU; reproducibility; FAIR data; Chile

1. Summary

Hospital emergency departments must absorb demand that changes with epidemiological conditions, season, age structure, and local service organization. Respiratory illness is particularly relevant because recurrent winter epidemics can concentrate consultations and hospitalizations within a limited number of weeks. Forecasting approaches have consequently been used to anticipate emergency attendance, bed demand, and severe respiratory hospitalizations [1–4]. Methodological work on emergency-department demand forecasting spans early systematic reviews of statistical models [5] through unified and multivariable forecasting pipelines for hourly and daily arrivals and occupancy [6–10]. Recent systematic reviews show both the growing use of machine-learning methods in emergency-demand prediction and persistent challenges related to external validation, generalizability, and operational integration [11,12]. Pathogen-specific forecasting systems have likewise targeted influenza, respiratory syncytial virus, and combined winter respiratory pressure on hospitals [13–16]. In Chile, empirical analyses have documented temporal variation in emergency consultations and pronounced respiratory-season effects [17,18]. Complementary Chilean clinical guidance and recent pediatric commentary describe the evolving respiratory-syncytial-virus and severe acute respiratory-infection context relevant to seasonal preparedness [19,20].

Within this setting, the Chilean Ministry of Health makes weekly respiratory emergency data publicly available through the Departamento de Estadísticas e Información de Salud (DEIS). The source dataset, produced from the Sistema de Atención Diaria de Urgencias (SADU), reports consultations and hospitalizations by respiratory cause, establishment, statistical week, and age group [21]. Although the source is highly valuable, downstream forecasting studies must repeatedly solve the same data-engineering tasks: retrieving a large Parquet file, harmonizing field names and types, isolating hospital emergency units, separating consultations from hospitalization rows, checking temporal coverage, pivoting causes to hospital-week features, constructing lagged predictors without leakage, and documenting provenance.

From a data-publication perspective, the principal contribution of Chile-ED-Resp is therefore not a new predictive algorithm. Instead, the resource provides a transparent and reusable research layer over the official source. The design follows the FAIR principles of findability, accessibility, interoperability, and reusability [22], including implementation practices such as machine-readable metadata, versioned provenance, and automated validation [23–28]. Table 1 now compares the release with the closest Chilean and Latin American emergency or respiratory resources rather than with datasets from unrelated domains.

Table 1: Closest Chilean and Latin American emergency or respiratory data resources and the distinct scope of Chile-ED-Resp.

Resource	Unit and scope	Preparation for reuse	Difference from Chile-ED-Resp
DEIS weekly respiratory-emergency source, Chile [21]	Establishment-week counts by respiratory cause and age group	Mutable official Parquet/XLSX resource and data dictionary	Direct upstream source; not versioned as a forecasting benchmark
DEIS public-network emergency data, Chile [29]	Annual emergency-attendance files derived from DAU records	Year-specific ZIP files and official dictionary	Broader emergency scope; no curated fixed hospital-week respiratory panel
SIVEP-Gripe, Brazil [30]	Case-level severe acute respiratory syndrome surveillance	Public annual micro-data and documentation	Clinical surveillance cases rather than hospital-level emergency-demand series
PAHO respiratory-virus open data [31]	Country/territory surveillance indicators for the Americas	Regional harmonized portal	Country-level surveillance rather than Chilean hospital-week demand
Chile-ED-Resp (this work)	180 hospitals on a fixed 209-week calendar	Versioned long and forecasting artifacts; checksums, audits, tests, and rolling-origin example	Reproducible hospital-level demand benchmark with explicit zero/missing semantics

Accordingly, Chile-ED-Resp releases two complementary representations of the same official 2022–2025 respiratory emergency information. The first is a curated long-format table that preserves all qualifying hospital emergency-unit records and official count values. The second is a hospital-week forecasting artifact that reorganizes selected consultation causes and adds documented deterministic features. An explicit coverage criterion identifies establishments suitable for benchmarking rather than hard-coding a small set of hospitals. The repository records dimensions, inventories, descriptive audits, source checksum, retrieval time, and validation results, allowing later versions to remain auditable if the upstream portal revises historical records.

2. Data Description

2.1. Official Source and Scope

The upstream resource is the government dataset *Atenciones de urgencias de causas respiratorias por semana epidemiológica*, published through `datos.gob.cl` by the Chilean Ministry of Health/DEIS [21]. The active Parquet resource has identifier `ae6c9887-106d-4e98-8875-40bf2b836041`. At the time this descriptor was prepared, the portal identified the resource under a Creative Commons Non-Commercial license; the linked terms correspond to Creative Commons Attribution-NonCommercial 2.0 Generic (CC BY-NC 2.0) [21,32]. The source field is named `SemanaEstadistica`; consequently, the manuscript uses *statistical week* for the analytical coordinate and retains *epidemiological week* only when reproducing the official Spanish dataset title. Because the current reporting year is continuously updated, the curated release intentionally uses the closed four-year interval 2022–2025. Restricting the versioned release to completed years avoids treating a partially reported current week as a genuine demand decline. This concern for completeness and timeliness mirrors broader open-government-data quality frameworks [33,34].

The source schema contains 25 variables after normalization to snake-case names. These fields describe establishment, region, municipality, health service, establishment type, administrative dependency, level of care, emergency-care type, coordinates, com-

plexity level, year, statistical week, cause code/order, cause label, total count, and five age-specific count fields. The official DEIS data dictionary documents the source schema and field definitions used to map the Parquet resource into the curated release [35].

Table 2: Variable families retained from the official DEIS/SADU source. The complete machine-readable dictionary is distributed as `metadata/data_dictionary.csv`.

Family	Main fields	Interpretation
Establishment	<code>establecimiento_codigo</code> , <code>establecimiento_glosa</code> , <code>tipo_establecimiento</code> , <code>nivel_atencion</code> , <code>nivel_complejidad</code>	Official identifiers and service characteristics.
Geography	<code>region_*</code> , <code>comuna_*</code> , <code>latitud</code> , <code>longitud</code>	Administrative and geographic descriptors.
Health service	<code>servicio_salud_*</code> , <code>dependencia_administrativa</code> , <code>tipo_urgencia</code>	Network and emergency-unit descriptors.
Time	<code>anio</code> , <code>semana_estadistica</code>	Official DEIS statistical year and week.
Respiratory event	<code>orden_causa</code> , <code>causa</code> , <code>num_total</code>	Cause label and total consultations or hospitalizations in the stratum.
Age strata	<code>num_menor1anio</code> , <code>num1a4anios</code> , <code>num5a14anios</code> , <code>num15a64anios</code> , <code>num65o_mas</code>	Official age-specific counts.

Table 3: Version 1.1.0 release snapshot generated from the official DEIS/SADU resource.

Release property	Value
Official source snapshot	Retrieved 30 August 2026 at 23:38:36 UTC; 3,629,980 rows; 67,929,078 bytes; resource ID <code>ae6c9887-106d-4e98-8875-40bf2b836041</code> .
Curated long artifact	449,844 rows $\times$ 32 variables; 180 hospitals; 12 official cause labels.
Forecasting artifact	37,620 rows $\times$ 38 variables; 180 hospitals $\times$ 209 expected statistical-week positions.
Benchmark coverage	178/180 hospitals (98.9%) satisfy $C_h \geq 0.95$; 172 have complete 209-week target coverage.
Validation	15 PASS, 0 WARN, 0 FAIL reported validation statuses.
Source integrity	SHA-256 recorded in <code>metadata/source_manifest.json</code> and fixed for the archived release.

2.2. Released Artifacts

Version 1.1.0 generates the artifacts summarized in Table 4. The long artifact contains 449,844 rows and 32 variables, whereas the forecasting artifact contains 37,620 rows and 38 variables. Both represent 180 hospitals; the long table retains 12 official respiratory cause labels. Exact dimensions and descriptive audits are stored with the release so that a mutable government source can be traced to the precise snapshot used for the build.

Table 4: Principal files created by the reproducible release pipeline.

File	Format	Role
data/chile_ed_resp_hospital_long.parquet	Parquet;	Official hospital emergency-unit records plus provenance and quality-control fields.
data/chile_ed_resp_hospital_long.csv.gz	CSV.gz	
data/chile_ed_resp_ira_alta_forecasting.parquet	Parquet; CSV	Fixed-grid hospital-week consultation features and deterministic predictors.
data/chile_ed_resp_ira_alta_forecasting.csv	CSV	
metadata/source_manifest.json	JSON	Retrieval time, identifiers, source URL, byte size, SHA-256, and license.
metadata/filter_audit.csv	CSV	Filtering counts and auditable cause-to-feature mapping.
metadata/cause_mapping.csv	CSV	Missing/zero frequencies and hospital-level coverage for every forecasting cause.
metadata/variable_completeness.csv	CSV	
metadata/forecast_variable_coverage.csv	CSV	
metadata/target_summary.csv	CSV	Distributional and week-level IRA Alta summaries.
metadata/temporal_summary.csv	CSV	
results/validation_checks.csv	CSV	Machine-readable structural and quality-control results, including records examined and flagged.
LICENSE-DATA	Text; YAML;	Redistribution terms, citation metadata, and permanent-archive deposit metadata.
CITATION.cff	JSON	
.zenodo.json		

2.3. Curated Long Table

The long table restricts the official source to the years 2022–2025, establishments classified as Hospital, and records whose emergency type contains the hospital-emergency designation UEH. Table 5 reports the row count after every filtering step. The final 449,844 rows represent 180 hospitals and 12 official respiratory cause labels; no cause is removed from the long artifact. Consultation codes 3–11 and hospitalization codes 33–35 are identified from the official `orden_causa/causa` combinations and stored in `measure_type`. This order-aware rule prevents hospitalization totals from entering consultation predictors.

Table 5: Sequential filtering audit for Version 1.1.0. Counts are also distributed in `metadata/filter_audit.csv`.

Step	Retained records	Rows retained	Rows removed at step
0	Official Parquet snapshot	3,629,980	—
1	Statistical years 2022–2025	1,547,292	2,082,688
2	<code>tipo_establecimiento = Hospital</code>	455,784	1,091,508
3	<code>tipo_urgencia</code> contains UEH	449,844	5,940

Beyond normalization of column names and coordinate types, official count fields are retained unchanged. Derived fields include a deterministic row identifier, `statistical_period`, the fixed source-resource identifier, the sum of the five official age strata, the difference between that sum and `num_total`, and a quality flag. All six official count variables are complete in the selected long table; their explicit-zero frequencies are reported in Table 7. Any future missing source value remains missing rather than being imputed. Such treatment supports transparent downstream handling of administrative health data [36].

2.4. Forecasting-Ready Hospital-Week Table

For reuse in time-series forecasting, consultation rows are mapped to canonical variables using the auditable rules in Table 6. The official label *IRA Alta (J00–J06)* is translated as *acute upper respiratory infection*; the Spanish label and acronym *IRA Alta* are preserved in the data and manuscript because they are the official source terminology. Aggregate consultation categories, crisis obstructiva bronquial, and all hospitalization rows are retained in the long artifact but excluded from the forecasting features. No source value is reinterpreted during the pivot.

Table 6: Exact cause-by-cause mapping from original DEIS/SADU labels to forecasting fields. Wording and capitalization reproduce the source; `metadata/cause_mapping.csv` preserves the verbatim strings, including spacing and punctuation.

ID	Official Spanish label	Measure	Forecast field
3	TOTAL CAUSA SISTEMA RESPIRATORIO (J00–J98)	Consult.	Excluded aggregate
4	IRA Alta (J00–J06)	Consult.	<code>ira_alta</code>
5	Influenza (J09–J11)	Consult.	<code>influenza</code>
6	Neumonía (J12–J18)	Consult.	<code>pneumonia</code>
7	Bronquitis/bronquiolitis aguda (J20–J21)	Consult.	<code>bronchitis_</code> <code>bronchiolitis</code>
8	Crisis obstructiva bronquial (J40–J46)	Consult.	Excluded non-target category
9	Otra causa respiratoria no contenidas en las categorías anteriores (J22; J30–J39, J47, J60–J98)	Consult.	<code>other_</code> <code>respiratory</code>
10	TOTAL ATENCIONES POR COVID-19 Virus no Identificado U07.2	Consult.	<code>covid19_</code> <code>unidentified</code>
11	TOTAL ATENCIONES POR COVID-19, Virus Identificado U07.1	Consult.	<code>covid19_</code> <code>identified</code>
33	- Causas sistema respiratorio (J00–J98)	Hosp.	Excluded hospitalization
34	- Por covid-19, virus no identificado U07.2	Hosp.	Excluded hospitalization
35	- Por covid-19, virus identificado U07.1	Hosp.	Excluded hospitalization

The mapping is order-aware rather than a fuzzy text substitution: only consultation orders 4, 5, 6, 7, 9, 10, and 11 receive canonical forecasting fields. Consequently, labels that contain similar respiratory or COVID-19 wording but belong to hospitalization orders 33–35 cannot enter the forecasting variables.

An absent hospital-week-cause record is interpreted as unavailable reporting, not as zero events. Explicit zeros are values for which the official source contains a record with `num_total = 0`. Accordingly, the pivot retains absent records as NA and preserves recorded zeros as 0; these states are never treated as interchangeable. Table 7 shows that every forecasting cause has 133 missing grid positions (0.35%), while zero frequencies vary materially by cause.

Mandatory interpretation rule. In the released tables, NA means that no qualifying source record was available for that hospital-week-cause combination, whereas 0 means that a qualifying official row explicitly reported zero events. Users must not replace the former with the latter unless an external reporting model justifies that recoding.

Table 7: Missingness and explicit-zero frequencies for released count variables. Zero percentages use non-missing observations as the denominator.

Artifact	Variable	Missing, n (%)	Explicit zero, n	Zero among observed (%)
Long	num_total	0 (0.00)	153,394	34.10
Long	num_menor1anio	0 (0.00)	320,982	71.35
Long	num1a4anios	0 (0.00)	280,005	62.24
Long	num5a14anios	0 (0.00)	263,106	58.49
Long	num15a64anios	0 (0.00)	199,422	44.33
Long	num65o_mas	0 (0.00)	222,460	49.45
Forecast	ira_alta	133 (0.35)	625	1.67
Forecast	influenza	133 (0.35)	16,091	42.92
Forecast	pneumonia	133 (0.35)	2,679	7.15
Forecast	bronchitis_	133 (0.35)	1,934	5.16
	bronchiolitis			
Forecast	other_respiratory	133 (0.35)	3,529	9.41
Forecast	covid19_	133 (0.35)	23,867	63.67
	unidentified			
Forecast	covid19_	133 (0.35)	23,274	62.09
	identified			

Hospital selection for benchmarking is coverage-based. The denominator is the fixed 209-position release calendar, not the set of weeks in which IRA Alta happens to appear somewhere nationally. Of the 180 hospitals, 172 report IRA Alta in all 209 positions; six additional hospitals remain above the 95% threshold with 204–208 observed target weeks. Consequently, 178 hospitals (98.9%) receive recommended_for_benchmark = True. The other two hospitals have 178/209 (85.2%) and 121/209 (57.9%) observed positions. Across the complete 180×209 grid, 37,487 of 37,620 IRA Alta values are observed (99.65% completeness). Within the recommended cohort, 37,188 of 37,202 values are observed (99.96%), leaving 14 missing targets. Table 8 summarizes the distribution.

Table 8: Distribution of IRA Alta temporal coverage C_h across the 180 hospitals in Version 1.1.0.

Coverage bin (C_h)	Hospitals	Share of 180	Benchmark status
209/209 target weeks (100%)	172	95.6%	Recommended
204–208/209 target weeks (97.6–99.5%)	6	3.3%	Recommended
178/209 and 121/209 target weeks (85.2% and 57.9%)	2	1.1%	Not recommended
Total (all establishments retained)	180	100%	178/180 (98.9%) recommended

Table 9 extends the hospital-level coverage audit to every forecasting cause. All seven variables share the same missingness mask because the selected official cause rows occur together at the retained hospital–week coordinates; their explicit-zero frequencies nevertheless differ, as shown in Table 7. The complete machine-readable audit is distributed as metadata/forecast_variable_coverage.csv.

Table 9: Missingness and hospital-level coverage for every forecasting cause on the $180 \times 209 = 37,620$ fixed grid.

Variable	Observed, n	Missing, n (%)	Complete hospitals (209/209)	Hospital coverage groups
ira_alta	37,487	133 (0.35)	172	6 with 204–208; 2 below 199
influenza	37,487	133 (0.35)	172	6 with 204–208; 2 below 199
pneumonia	37,487	133 (0.35)	172	6 with 204–208; 2 below 199
bronchitis_ bronchiolitis	37,487	133 (0.35)	172	6 with 204–208; 2 below 199
other_respiratory	37,487	133 (0.35)	172	6 with 204–208; 2 below 199
covid19_ unidentified	37,487	133 (0.35)	172	6 with 204–208; 2 below 199
covid19_ identified	37,487	133 (0.35)	172	6 with 204–208; 2 below 199

The 180 hospitals cover all 16 Chilean regions (Table 10). Hospital-week meta-data are joined on hospital, statistical year, and statistical week, so a latest record is no longer propagated backward. A separate stability audit found no within-hospital changes across the 12 released metadata fields during 2022–2025, but the time-specific implementation avoids relying on that empirical stability.

Table 10: Hospitals by region in Version 1.1.0.

Region	Hospitals	Region	Hospitals
Metropolitana de Santiago	24	La Araucanía	23
Los Lagos	20	Biobío	20
Valparaíso	19	O’Higgins	15
Maule	13	Coquimbo	9
Los Ríos	9	Ñuble	7
Antofagasta	5	Atacama	5
Aysén	5	Magallanes	4
Arica y Parinacota	1	Tarapacá	1

IRA Alta has 37,487 observed hospital-week values, a mean of 76.63 (SD 73.14), median 58 (interquartile range 29–102), range 0–4,415, and aggregate total 2,872,473. Table 11 shows the year-specific distribution and national weekly peak. The peak shifts between statistical weeks 20 and 43 across years, illustrating the temporal heterogeneity that motivates year-aware validation.

Table 11: Distribution and temporal summary of acute upper respiratory infection (official label: IRA Alta).

Period	Observed missing	Zeros	Mean SD	Median IQR	Peak week (total)
2022	9,352 / 8	163	86.38 $\pm$ 95.13	61 [28–116]	W43 (26,615)
2023	9,298 / 62	151	80.11 $\pm$ 70.31	61 [32–106]	W20 (26,279)
2024	9,307 / 53	161	75.24 $\pm$ 63.39	59 [33–101]	W20 (26,031)
2025	9,530 / 10	150	65.01 $\pm$ 56.35	51 [26–88]	W43 (16,280)
Overall	37,487 / 133	625	76.63 $\pm$ 73.14	58 [29–102]	2022–W43 (26,615)

Temporal features are constructed only after sorting by hospital and continuous `time_index`. The release provides one-, two-, and four-week lags, `ira_alta_prev_year_same_week`, trailing four- and 12-week means, and within-year sine/cosine encodings. A chronological split labels 2022–2024 as training and 2025 as testing. The information available for a one-week-ahead origin t comprises target observations through $t - 1$; therefore, the stored target-derived predictors support rolling-origin one-step evaluation. They must not be used unchanged for a fixed-origin multi-step forecast of all 2025, because later 2025 rows incorporate outcomes observed during 2025. For fixed-origin evaluation, users must generate predictors recursively or restrict models to information available at the fixed origin [37].

3. Methods

3.1. Reproducible Processing Workflow

Figure 1 summarizes the curation workflow. The process begins from the stable government resource identifier rather than from a manually downloaded local copy. For Version 1.1.0, the official Parquet snapshot was retrieved on 30 August 2026 at 23:38:36 UTC, contained 3,629,980 rows, had a size of 67,929,078 bytes, and produced the SHA-256 digest reported in the source manifest and Data Availability Statement. The downloader verifies the Parquet signature, computes the digest, and writes provenance before transformation. The raw government file is excluded from Git version control; the release stores its manifest and redistributes only the curated 2022–2025 artifacts under the source-compatible license.

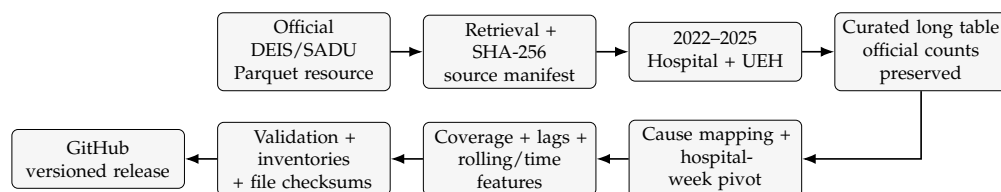


Figure 1. Reproducible Chile-ED-Resp curation workflow. All demand counts originate in the official DEIS/SADU resource; the pipeline adds deterministic organizational, provenance, quality-control, and forecasting fields.

3.2. Schema Harmonization and Type Treatment

Column names are normalized to lower-case snake case. Because public-data schemas can appear with either CamelCase or previously cleaned headings, the build script matches fields through an underscore-insensitive canonical key before checking

for the complete expected 25-field schema. If an expected source field is absent, the build stops rather than guessing a replacement. Numeric count, year, week, and cause-order columns are explicitly parsed; coordinates are converted from decimal-comma text to numeric decimal-point representation. No other source values are standardized in a way that changes their substantive meaning.

The curation script also keeps an explicit cause inventory. Consultation versus hospitalization is assigned from the official cause-order group, and `canonical_feature` is assigned only to consultation orders 4, 5, 6, 7, 9, 10, and 11. The inventory records every official label, measure class, number of source rows, aggregate reported total, canonical field, inclusion status, and exclusion reason. Maintaining the original label alongside canonical features allows users to audit the mapping and rebuild alternative targets.

3.3. Fixed Statistical-Week Calendar and Time Index

The release calendar is constructed before inspecting hospital-specific target availability. The official DEIS coordinates present in the archived 2022–2025 source define weeks 1–52 for 2022, 2023, and 2024 and weeks 1–53 for 2025, yielding 209 expected positions. The build checks the observed national IRA Alta coordinate set against this complete grid, validates the week-53 transition, and crosses all 180 hospitals with the grid. A zero-based `time_index` runs continuously from 0 to 208, and `weeks_in_statistical_year` records whether the corresponding year contains 52 or 53 positions. Consequently, a week absent everywhere in a future source snapshot would remain an explicit all-missing grid position rather than causing later row-shift lags to cease representing consecutive calendar positions.

3.4. Hospital Coverage and Selection

Rather than selecting hospitals on the basis of peak demand, geographic convenience, or prior model performance, the release applies a transparent completeness criterion. Let W^* denote the fixed set of 209 expected statistical-week positions and let $W_h \subseteq W^*$ denote positions with an observed IRA Alta value for hospital h . Coverage is

$$C_h = \frac{|W_h|}{|W^*|} = \frac{|W_h|}{209}. \quad (1)$$

The fixed denominator prevents a nationally absent source record from inflating every hospital's coverage. The default benchmark recommendation is $C_h \geq 0.95$. On the 209-position calendar, this rule requires at least 199 observed target weeks and permits at most 10 missing weeks (4.8%). It was specified as a completeness convention independently of forecasting outcomes. The observed coverage distribution provides an additional empirical check: the least-complete retained hospital has 204/209 weeks (97.6%), whereas the most-complete excluded hospital has 178/209 weeks (85.2%). Thus, the 0.95 cutoff lies inside an observed separation gap, and any threshold in $(178/209, 204/209]$ yields the same 178-hospital benchmark cohort. This threshold is not a clinical quality criterion or a claim that 0.95 is universally optimal; it limits avoidable missingness while retaining 98.9% of hospitals, and the full 180-hospital grid remains available for sensitivity analyses using alternative criteria.

3.5. Feature Engineering Without Leakage

For a hospital h and target series $y_{h,t}$ ordered by `time_index`, short-term lag features are $y_{h,t-k}$ for $k \in \{1, 2, 4\}$. The year-over-year feature is not a fixed 52-row shift. For statistical year a and statistical week w , it matches $y_{h,a,w}^{(\text{prev-year})} = y_{h,a-1,w}$ and is therefore named `ira_alta_prev_year_same_week`. This definition handles the 2025

week-53 transition explicitly; where the preceding year lacks week 53 or the matched source value is unavailable, the feature is missing. Prior rolling means use

$$\bar{y}_{h,t}^{(m)} = \frac{1}{m} \sum_{j=1}^m y_{h,t-j}, \quad m \in \{4, 12\}, \quad (2)$$

whenever all required prior values exist. Because every hospital is crossed with the complete calendar before feature computation, $t - k$ denotes the exact preceding grid position rather than the k th preceding observed value. Thus, a missing $y_{h,t-k}$ produces a missing lag and is never bypassed. Rolling windows likewise operate on the preceding four or 12 grid positions with the required observation count equal to the window length; if any position in that window is missing, the rolling mean remains missing rather than shortening the window or skipping the gap. Boundary positions without sufficient history also remain missing. The current observation $y_{h,t}$ is excluded from each rolling feature.

Seasonal position is encoded as

$$s_t = \sin\left(\frac{2\pi(w_t - 1)}{m_t}\right), \quad c_t = \cos\left(\frac{2\pi(w_t - 1)}{m_t}\right), \quad (3)$$

where w_t is the statistical-week number and $m_t \in \{52, 53\}$ is the number of positions in that statistical year. In a 53-week statistical year, week 53 is retained as a distinct observation and is encoded at angle $2\pi(52/53)$, the last unique phase before week 1 of the next year returns to $(s_t, c_t) = (0, 1)$. It is neither dropped nor duplicated as week 1. In a 52-week year, the denominator is 52; consequently, the same numbered week can have a slightly different normalized annual phase across 52- and 53-week years by design. Therefore, the implementation does not use a fixed denominator of 52: it reads `weeks_in_statistical_year` for every row. The information set for rolling-origin one-week-ahead evaluation at origin t is $\mathcal{I}_{h,t} = \{y_{h,s} : s < t\}$ together with non-target covariates available by t . The repository example reconstructs all target-derived predictors from $\mathcal{I}_{h,t}$ at every origin and appends $y_{h,t}$ to the history only after the forecast has been scored. For a fixed-origin horizon $H > 1$, users must recursively update target-derived features or omit predictors unavailable at the origin; the precomputed 2025 columns do not by themselves establish leakage-free fixed-origin evaluation.

3.6. Automated Build and Versioning

The public repository contains Python scripts for source retrieval, curation, validation, descriptive audits, and rolling-origin evaluation. Unit tests cover schema normalization, cause/measure separation, the 209-position calendar, and continuous time indexing. The GitHub Actions workflow uses Python 3.12.13 and installs `requirements.lock` with `-require-hashes`; this file fixes the complete direct and transitive dependency graph to exact versions and verifies package hashes. The human-maintained `requirements.txt` retains only development-level minimum constraints and is not used for the reproducible workflow. The action then executes the tests, downloads the official resource, builds both data artifacts, runs validation, and writes the versioned outputs and metadata. The evaluation script implements a one-week horizon for 2025 and recomputes lag-1, lag-2, lag-4, four- and 12-week rolling means, and the previous-year same-week value at every forecast origin without reading the precomputed predictor columns. This approach makes the publication package independently rebuildable without distributing the upstream raw file [38,39].

Version-specific provenance is strengthened through cryptographic hashes. The source manifest records retrieval time, file size, and SHA-256 of the official Parquet snapshot; the final checksum file records hashes of the curated Parquet/CSV artifacts and principal metadata files. Similar checksum-based provenance has recently been

used in a national health-data Data Descriptor to distinguish an immutable research release from mutable upstream public files [40].

3.7. Technical Validation

Validation is designed to detect structural and source-quality issues without rewriting official observations. Table 12 lists the automated checks. A FAIL status stops the release when the forecasting table cannot be trusted structurally; a WARN status exposes a potentially relevant anomaly while preserving the source values for transparent secondary analysis.

Table 12: Validation controls and record counts for the 30 August 2026 Version 1.1.0 build. The same fields are distributed in `results/validation_checks.csv`.

Check	Records examined	Observed result	Status
Official source counts preserved	2,699,064 count cells	0 modified cells	PASS
Analysis years	449,844 long rows	0 rows outside 2022–2025	PASS
Non-negative counts	2,699,064 count cells	0 negative cells	PASS
Age-total reconciliation	449,844 long rows	0 mismatches	PASS
Unavailable age reconciliation	449,844 long rows	0 unavailable checks	PASS
Duplicate source strata	449,844 long rows	0 duplicated-key rows	PASS
Unique hospital-week forecast key	37,620 forecast rows	0 duplicated keys	PASS
Chronological ordering	37,620 forecast rows	0 order violations	PASS
Chronological split	37,620 forecast rows	0 invalid split rows	PASS
Fixed 209-week calendar	209 national positions	0 missing/unexpected positions	PASS
Continuous time index	209 positions	0 discontinuities	PASS
Complete hospital-calendar grid	37,620 expected rows	0 missing/extra rows	PASS
Consultation–hospitalization separation	449,844 long rows	0 misclassified rows	PASS
Zero/missing state separation	37,620 target cells	133 missing; 625 explicit zeros; 0 conflations	PASS
No count imputation	2,699,064 count cells	0 imputed cells	PASS

For the 30 August 2026 build of Version 1.1.0, the validation report recorded 15 PASS, 0 WARN, and 0 FAIL statuses. Table 12 reports the exact scope of each check and retains zero-valued exception counts rather than replacing them with a generic pass indicator. It confirms 449,844 long-format rows, 37,620 forecasting rows, 180 hospitals, 12 official labels, the fixed 209-position grid, a continuous time index, and order-aware separation of consultation and hospitalization measures. The rolling-origin example writes origin-level predictions and establishment-level MAE, RMSE, and MAPE; these outputs demonstrate the evaluation protocol and are not presented as evidence that one forecasting model is superior.

4. User Notes

4.1. Reproducing the Release

Users can reproduce the complete release from a clean checkout with the commands below. The reproducible workflow uses Python 3.12.13 and the hash-verified `requirements.lock`; `requirements.txt` remains the editable declaration of minimum development requirements rather than the environment specification used for the release.

```
git clone https://github.com/cvidalmsu/chile-ed-resp-benchmark.git
cd chile-ed-resp-benchmark
python -m venv .venv
# activate the environment for the operating system in use
python -m pip install --require-hashes -r requirements.lock
pytest -q
python scripts/make_release.py --start-year 2022 --end-year 2025 --force-download
```

Alternatively, repository maintainers can run *Actions* → *Build DEIS-SADU dataset* → *Run workflow*. The action records the newly retrieved source checksum before rebuilding the release. For reuse and citation, users should reference the permanent Chile-ED-Resp Version 1.1.0 archive at <https://doi.org/10.5281/zenodo.22209795>, report the dataset version used in their analysis, and consult the public development repository at <https://github.com/cvidalmsu/chile-ed-resp-benchmark> for subsequent updates.

4.2. Recommended Forecasting Use

For comparable model experiments, the recommended starting point is the forecasting artifact:

```
data/chile_ed_resp_ira_alta_forecasting.parquet
```

Users seeking a high-completeness benchmark should initially restrict the table to the 178 establishments carrying the benchmark recommendation flag. The supplied `split` field provides a 2022–2024/2025 partition, but the valid interpretation of target-derived predictors depends on forecast design. For rolling-origin one-week-ahead evaluation, run `scripts/rolling_origin_example.py`; the script reconstructs the information set at each origin and exposes each recomputed predictor in `results/rolling_origin_predictions.csv`. For a fixed origin preceding 2025, later 2025 target-derived columns must not be used as known inputs; generate them recursively or select origin-available covariates. Randomly shuffled weekly observations are unsuitable for either design [37]. High-dimensional models should be interpreted cautiously because model complexity and sample frequency can materially affect performance in emergency-demand forecasting [11,41].

For descriptive analyses, the long table is preferable because it retains official age strata, cause wording, hospitalizations, and establishment metadata. Users should avoid summing consultation and hospitalization rows into a single outcome unless the analytical question explicitly requires such an aggregate. The `measure_type`, `causa`, and `orden_causa` fields make that distinction inspectable. Likewise, absent records must not be recoded as zeros without a justified external reporting model.

4.3. Licensing, Attribution, and Provenance

The official resource page labels the source under a Creative Commons Non-Commercial license [21]; the release documents the corresponding CC BY-NC 2.0 licensing terms [32]. The curated tables are adaptations of those records and may be copied, shared, and further adapted only for non-commercial purposes under those terms. Redistribution must retain reasonable attribution to the Chilean Ministry of Health/DEIS, the official dataset title and source link, the CC BY-NC 2.0 license URI, and applicable license/disclaimer notices; it must also identify the transformations made by Chile-ED-Resp and any further changes made by the redistributor. Redistributors may not add legal or technological restrictions that prevent recipients from

exercising the licensed rights. Use primarily intended for commercial advantage or private monetary compensation requires separate permission from the applicable rights holder. The repository file LICENSE-DATA provides a practical redistribution checklist and the authoritative license link. The MIT license applies only to the software and does not override the non-commercial condition attached to the source-derived data; the original DEIS/SADU records are not relicensed by the authors.

4.4. Limitations and Reuse Hazards

Several limitations should guide interpretation. First, the resource contains aggregate administrative counts rather than individual patient records; it cannot support patient-level risk prediction, causal inference, or analyses requiring comorbidity trajectories. Second, the meaning and completeness of administrative reporting depend on the upstream SADU system and local establishment practices. The pipeline reports anomalies but cannot independently verify whether every source count reflects complete clinical activity. Third, retrospective corrections by the public agency can change the current upstream file; source manifests and versioned releases are therefore essential when reproducing a published analysis.

Further limitations arise from temporal and modeling choices. The 2022–2025 window was selected to create a closed contemporary period, not to claim that earlier years are unusable. The 95% IRA Alta coverage threshold is a reproducible benchmarking rule rather than a universal hospital-inclusion standard. Derived lags and rolling means remain missing when history is insufficient or when an exact required grid position is unavailable; gaps are never skipped or collapsed. Moreover, precomputed target-derived fields encode observations through the preceding row and therefore support sequential one-week origins, not a fixed-origin 53-week claim. No weather, staffing, bed-occupancy, holiday, or socioeconomic variable is added to the core release because such variables require separate provenance and licensing documentation.

Finally, the data should not be interpreted as a direct measure of latent population disease incidence. Emergency consultations reflect both morbidity and healthcare-seeking behavior, local service organization, referral patterns, and reporting practice. Forecasting models built from the resource are therefore best understood as models of reported emergency demand within the represented public-service context. Table 13 summarizes these limitations alongside recommended mitigations for downstream users.

5. Conclusions

In summary, Chile-ED-Resp converts a valuable but operationally oriented Chilean public-health resource into a documented and reproducible research dataset without replacing official count observations. Version 1.1.0 comprises a 449,844-row, 32-variable long artifact and a 37,620-row, 38-variable hospital-week artifact covering 180 hospitals, 12 official labels, and 209 expected statistical weeks during 2022–2025. A fixed-denominator coverage rule identifies 178 hospitals for benchmarking while retaining all 180 establishments. The release provides a verbatim cause-to-variable mapping, reports coverage for all seven forecasting causes, distinguishes missing records from explicit zeros, preserves time-specific metadata, excludes hospitalization codes from consultation features, and provides a one-week rolling-origin example. Fifteen validation checks pass with no warnings or failures, and their record scopes and zero-valued exception counts are reported explicitly. Source manifests, checksums, descriptive audits, tests, licensing guidance, and automated rebuilding support traceability from DEIS/SADU to the versioned release.

Author Contributions: Conceptualization, W.B.-Q., C.A. and C.V.-S.; methodology, C.A., C.C.-G., B.S.-N., J.F.-B. and C.V.-S.; software, B.S.-N., J.F.-B. and M.T.-Y.; validation, C.C.-G., B.S.-N., J.F.-B. and M.T.-Y.; formal analysis, C.A., C.C.-G., B.S.-N. and J.F.-B.; data curation, B.S.-N., J.F.-B. and C.C.-G.; writing—original draft preparation, W.B.-Q., C.A., B.S.-N. and J.F.-B.; writing—review

Table 13: Principal limitations of Chile-ED-Resp and recommended mitigations for reuse.

Limitation	Recommended mitigation / user guidance
Aggregate administrative counts, not patient-level records	Do not use for patient-level risk prediction or causal inference; restrict to hospital- or population-level demand analysis
SADU reporting completeness depends on the upstream system and local establishment practice	Prioritize the 178 hospitals flagged <code>recommended_for_benchmark = True</code> ; consult <code>hospital_coverage.csv</code> before pooling establishments
Upstream DEIS/SADU file may be retrospectively corrected by the public agency	Cite the Chile-ED-Resp GitHub repository, report the dataset version used, and preserve the SHA-256 checksum of the corresponding source snapshot rather than relying only on the mutable upstream portal URL
The 95% coverage threshold is a reproducibility convention, not a clinical quality standard	Recompute C_h with an alternative threshold using the complete 180-hospital grid if a different inclusion rule is required
Lags and rolling means are missing where history is insufficient or an exact required grid position is unavailable	Preserve the fixed time axis; do not skip gaps or substitute zero for absence
Target-derived 2025 fields become available sequentially	Use the rolling-origin script for one-week forecasts; reconstruct recursively for fixed-origin multi-week evaluation
Absent hospital-week-cause record is not an event count	Preserve as missing; use only explicit source records with <code>num_total = 0</code> as zero events
Consultations reflect healthcare-seeking behavior as well as morbidity	Interpret forecasts as models of reported emergency demand, not of latent disease incidence

and editing, all authors; visualization, B.S.-N., J.F.-B. and M.T.-Y.; supervision, C.A. and C.V.-S.; project administration, W.B.-Q. and C.V.-S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable. The dataset is derived from publicly available, aggregate administrative health statistics and contains no patient-level records or direct personal identifiers.

Informed Consent Statement: Not applicable. No individual participants were recruited and no identifiable individual-level data were accessed.

Data Availability Statement: The curated dataset, processing code, machine-readable descriptive audits, validation outputs, and documentation for Version 1.1.0 are permanently archived in Zenodo at <https://doi.org/10.5281/zenodo.22209795>. The public development repository is available at <https://github.com/cvidalmsu/chile-ed-resp-benchmark>. Version 1.1.0 was built from the official DEIS/SADU Parquet snapshot retrieved on 30 August 2026 at 23:38:36 UTC (67,929,078 bytes). The source SHA-256 is `ef21c621527091dc109ecd8a66f11199d8680b50435d4d77cf69626a08d90e0`. The upstream records are publicly available from the Chilean Ministry of Health/DEIS dataset *Atenciones de urgencias de causas respiratorias por semana epidemiológica*, resource ID `ae6c9887-106d-4e98-8875-40bf2b836041`.

Acknowledgments: The authors acknowledge the Departamento de Estadísticas e Información de Salud (DEIS), Ministerio de Salud de Chile, for maintaining the public SADU respiratory-emergency dataset used as the source of this curated resource.

Conflicts of Interest: The authors declare no conflicts of interest.

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